

## FILE HANDLING IN C

### 71. Writing and Reading Characters using `putc()` and `getc()`

Write a C program that:

1. Opens a file `test.txt` in write mode.
2. Writes the word "HELLO" character by character using `putc()`.
3. Closes the file.
4. Reopens the file in read mode.
5. Reads the contents using `getc()` and displays them on the screen.

### 72. Counting Characters using EOF

A file `data.txt` contains:

Welcome  
To  
C Programming

Write a program that:

1. Opens the file in read mode.
2. Reads characters using `getc()`.
3. Uses EOF to terminate reading.
4. Counts and displays the total number of characters (including spaces and newline characters).

### 73. Writing and Reading Integer Array using putw() and getw()

Given:

```
int arr[] = {100, 200, 300, 400, 500};
```

Write a program to:

1. Write all array elements to `numbers.txt` using `putw()`.
2. Read them back using `getw()`.
3. Display all numbers.
4. Calculate and display their sum.

### 74. Writing Formatted Data using fprintf()

Student details:

- Name = "Alice"
- Roll No = 10
- Marks = 87.5

Write a program that:

1. Writes this data into `student.txt` using `fprintf()`.
2. Stores the return value of `fprintf()`.
3. Displays the number of characters written into the file.

## 75. Reading Multiple Records using fscanf()

The file `records.txt` contains:

```
Amit 20 85.5  
Neha 21 90.0  
Rahul 19 88.0
```

Write a program that:

1. Reads all records using `fscanf()` inside a loop.
2. Displays each record.
3. Finds and displays the student with the highest marks.

## 76. File Opening Error Handling

Write a program that:

1. Attempts to open `unknown.txt` in read mode.
2. Checks whether the file pointer is `NULL`.
3. Displays an appropriate error message if the file cannot be opened.

## 77. Using feof() and ferror()

Write a program that:

1. Opens a file in read mode.
2. Reads data character by character.
3. Uses `feof()` to detect end-of-file.

4. Uses `ferror()` to detect if an error has occurred.
5. Displays appropriate messages.

## 78. Demonstrating `ftell()` and `rewind()`

Write a program that:

1. Writes some text into a file.
2. Uses `ftell()` to display the current file position.
3. Uses `rewind()` to move the file pointer to the beginning.
4. Again uses `ftell()` and displays the new position.
5. Explain the difference in positions.

## 79. Random Access using `fseek()`

A file `alpha.txt` contains:

ABCDEFGHIJ

Write a program that:

1. Opens the file in read mode.
2. Uses `fseek(fp, 5, SEEK_SET)` to move the file pointer.
3. Reads and displays the character at that position.

## 80. Comprehensive File Handling Program

The file `accounts.txt` contains:

```
1001 Ravi 5000
1002 Neha 7000
1003 Amit 4500
```

Write a program that:

1. Reads all records using `fscanf()`.
2. Calculates the total balance.
3. Finds the account with the highest balance.
4. Uses `fseek()` to directly access the second record.
5. Uses `ftell()` to display the current file position.
6. Uses `feof()` to check end-of-file.
7. Uses `ferror()` to detect any file error.